LESSON 9
BAR NUMBERS

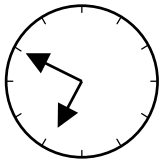
SUMMARY

9.1 **Removing Bar Numbers** – converting numbers containing a negative digit to positive form.
9.2 **Subtraction** – a general subtraction method.
9.3 **Creating Bar Numbers** – removing digits over 5 from a number.
9.4 **Using Bar Numbers** – some applications of bar numbers.

9.1 REMOVING BAR NUMBERS

The number 19 is very close to 20.
And it can therefore be conveniently written in a different way: as $2\bar{1}$

$2\bar{1}$ means 20 – 1, the minus is put on top of the 1.
Similarly $3\bar{1}$ means 30 – 1 or 29.
And $4\bar{2}$ means 38.
This is like telling the time when we say 'ten to seven' or instead of 6:50.



We pronounce $4\bar{2}$ as "four, bar two" because the 2 has a bar on top.

1 **$7\bar{2}$ = 68**,

$86\bar{1}$ = 859, because $6\bar{1}$ = 59 (the 8 is unchanged),

$127\bar{2}$ = 1268, because $7\bar{2}$ = 68,

$6\bar{3}0$ = 570, because we have 600 – 30 (or because $6\bar{3}$ = 57).

Practice A Convert the following numbers:

a $6\bar{1}$ **b** $8\bar{2}$ **c** $3\bar{3}$ **d** $5\bar{7}$ **e** $46\bar{2}$

f $999\bar{1}$ **g** $1\bar{2}$ **h** $11\bar{1}$ **i** $12\bar{3}$ **j** $3\bar{4}0$

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a 59 b 78 c 27 d 43 e 458
f 9989 g 8 h 109 i 117 j 260

Any digit in a number may have a bar on it.

2 How would you remove the bar number in $\mathbf{5\bar{1}3}$?

The best way is to split the number into two parts: $5\bar{1}/3$.
Since $5\bar{1} = 49$, the answer is **493**.

If a number has a bar number in it split the number after the bar.

3 $\mathbf{7\bar{3}1} = 7\bar{3}/1 = \mathbf{671}$,

$\mathbf{52\bar{4}2} = 52\bar{4}/2 = \mathbf{5162}$,

$\mathbf{3\bar{2}15} = 3\bar{2}/15 = \mathbf{2815}$ since $3\bar{2} = 28$,

$\mathbf{5\bar{1}3\bar{2}} = 5\bar{1}/3\bar{2} = \mathbf{4928}$ since $5\bar{1} = 49$ and $3\bar{2} = 28$,

$\mathbf{3\bar{1}3\bar{2}3\bar{3}} = 3\bar{1}/3\bar{2}/3\bar{3} = \mathbf{292827}$.

✐ **Practice B** Remove the bar numbers:

a $6\bar{1}4$	**b** $4\bar{2}3$	**c** $5\bar{2}5$	**d** $3\bar{1}7$
e $45\bar{2}3$	**f** $333\bar{2}3$	**g** $5\bar{1}32$	**h** $6\bar{2}7\bar{3}$
i $2\bar{1}1$	**j** $41\bar{3}1$	**k** $1\bar{3}15\bar{1}$	**l** $1\bar{3}1$

a 594	**b 383**	**c 485**	**d 297**
e 4483	**f 33283**	**g 4932**	**h 5867**
i 191	**j 4071**	**k 7149**	**l 71**

Next suppose the bar spans more than one digit in a number.

ALL FROM 9 AND THE LAST FROM 10

So far we have only had a bar on a single figure.
But we could have two or more bar numbers together.

4 Remove the bar numbers in $\mathbf{5\overline{33}}$.

The 5 means 500, and $\overline{33}$ means 33 is to be subtracted.
So $5\overline{33}$ means 500 – 33, and we have met sums like this in Lesson 5.

500 – 33 = **467** because the 33 comes off one of the hundreds, so the 5 is reduced to 4.

And applying *All from 9 and the Last from 10* to 33 gives 67.

5 Similarly $\mathbf{7\overline{14}}$ = **686** the 7 reduces to 6 and the Sutra converts 14 to 86,

$\mathbf{26\overline{21}}$ = **2579** 26 reduces to 25,

$\mathbf{7\overline{02}}$ = **698** the Sutra converts 02 to 98,

$\mathbf{50\overline{3}}$ = **497** 50 is reduced to 49 (alternatively, write $50\overline{3}$ as $5\overline{03}$: see previous example),

$\mathbf{4\overline{20}} = 4\overline{2}0$ = **380**.

6 $\mathbf{4\overline{23}1}$ = **3771**.

Here we can split the number after the bar: $4\overline{23}/1$.
$4\overline{23}$ changes to 377, and we just put the 1 on the end: $4\overline{23}1$ = **3771**.

7 Similarly $\mathbf{5\overline{12}4} = 5\overline{12}/4$ = **4884**,

$\mathbf{3\overline{11}33} = 3\overline{11}/33$ = **28933**,

$\mathbf{5\overline{123}}$ = **4877**,

$\mathbf{3\overline{1}4\overline{31}} = 3\overline{1}/4\overline{31}$ = **29369**.

✐ **Practice C** Remove the bar numbers:

a $6\overline{12}$ **b** $7\overline{33}$ **c** $5\overline{11}$ **d** $9\overline{04}$ **e** $72\overline{41}$ **f** $333\overline{22}$

g $6\overline{21}4$ **h** $5\overline{31}22$ **i** $33\overline{22}44$ **j** $7\overline{333}$ **k** $5\overline{104}$ **l** $44\overline{112}$

m $74\overline{031}$ **n** $7\overline{103}1$ **o** $6\overline{3322}$ **p** $3\bar{1}10\bar{2}$ **q** $3\overline{1}14\bar{1}$ **r** $3\bar{2}1\overline{22}$

a	**588**	**b**	**667**	**c**	**489**	**d**	**896**	**e**	**7159**	**f**	**33278**
g	**5794**	**h**	**46922**	**i**	**327844**	**j**	**6667**	**k**	**4896**	**l**	**43888**
m	**73969**	**n**	**68971**	**o**	**56678**	**p**	**29098**	**q**	**28939**	**r**	**28078**

ADVANTAGES OF BAR NUMBERS

Bar numbers are an ingenious device which we will be using in later work. Their main advantages are:

1. They give us flexibility: we use the vinculum when it suits us.
2. Large numbers, like 6, 7, 8, 9 can be avoided.
3. Figures tend to cancel each other, or can be made to cancel.
4. 0 and 1 occur twice as frequently as they otherwise would.

9.2 SUBTRACTION

These bar numbers give us an alternative way of subtracting numbers.
Pupils sometimes subtract in each column in a subtraction sum regardless of whether the top is greater than the bottom or not.
This method can however be used to give the correct answer.

8

$$\begin{array}{r} 444 \\ \underline{286} \; - \\ \end{array}$$

Subtracting in each column we get 4–2 = 2, 4–8 = –4, 4–6 = –2.
Since these negative answers can be written with a bar on top we can write:

$$\begin{array}{r} 444 \\ \underline{286} \; - \\ \underline{2\bar{4}\bar{2}} \end{array}$$

and $2\overline{42}$ is easily converted into **158**.

9 Similarly

$$\begin{array}{r} 6767 \\ \underline{1908} \; - \\ \underline{5\bar{2}6\bar{1}} \end{array} = \mathbf{4859}$$

✐ **Practice D** Subtract using bar numbers:

a 543
168 –
___

b 567
279 –
___

c 804
388 –
___

d 737
558 –
___

e 6413
1878 –
_____

f 8024
5339 –
_____

g 6543
2881 –
_____

h 7103
3991 –
_____

i 4545
1791 –
_____

j 3204
2081 –
_____

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a 375 **b 288** **c 416** **d 179** **e 4535**
f 2685 **g 3662** **h 3112** **i 2754** **j 1123**

9.3 CREATING BAR NUMBERS

We may also need to put numbers **into** bar form.

10

79 = 8$\bar{1}$ because 79 is 1 less than 80,

239 = 24$\bar{1}$ because 39 = 4$\bar{1}$,

7689 = 769$\bar{1}$ because 89 = 9$\bar{1}$.

508 = 51$\bar{2}$ 08 becomes 1$\bar{2}$

✐ **Practice E** Put the following into bar form:

a 49 **b** 58 **c** 77 **d** 88

e 69 **f** 36 **g** 17 **h** 359

i 848 **j** 7719 **k** 328 **l** 33339

m 609 **n** 708

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a 5$\bar{1}$ **b 6$\bar{2}$** **c 8$\bar{3}$** **d 9$\bar{2}$**
e 7$\bar{1}$ **f 4$\bar{4}$** **g 2$\bar{3}$** **h 36$\bar{1}$**
i 85$\bar{2}$ **j 772$\bar{1}$** **k 33$\bar{2}$** **l 3334$\bar{1}$**
m 61$\bar{1}$ **n 71$\bar{2}$**

One of the main advantages of bar numbers is that we can remove high digits in a number. For example writing **19** as $\mathbf{2\bar{1}}$ means we do not have to deal with the large 9.

11 Remove the large digits from **287**.

Here the 8 and the 7 are large (we say that 6, 7, 8, 9 are large digits).

So we write 287 as $\mathbf{3\overline{13}}$ the 2 at the beginning is increased to 3, and the Sutra *All from 9 and the Last from 10* is applied to 87 to give 13.

You will agree that 287 is 13 below 300, which is what $\mathbf{3\overline{13}}$ says.

12 Similarly $\mathbf{479 = 5\overline{21}}$,

$\mathbf{3888 = 4\overline{112}}$,

$\mathbf{292 = 3\bar{1}2}$,

$\mathbf{4884 = 5\overline{12}4}$,

$\mathbf{77 = 1\overline{23}}$ (you can think of 77 as 077),

and so on.

Practice F Remove the large digits from the following:

a 38	**b** 388	**c** 298	**d** 378
e 3991	**f** 3822	**g** 4944	**h** 390
i 299	**j** 98	**k** 87	**l** 888
m 996	**n** 2939	**o** 1849	**p** 7

a $\mathbf{4\bar{2}}$	**b** $\mathbf{4\overline{12}}$	**c** $\mathbf{30\bar{2}}$	**d** $\mathbf{4\overline{22}}$
e $\mathbf{4\overline{011}}$	**f** $\mathbf{4\bar{2}22}$	**g** $\mathbf{5\bar{1}44}$	**h** $\mathbf{4\bar{1}0}$
i $\mathbf{30\overline{1}}$ **or** $\mathbf{30\bar{1}}$	**j** $\mathbf{10\overline{2}}$	**k** $\mathbf{1\overline{13}}$	**l** $\mathbf{1\overline{112}}$
m $\mathbf{\overline{1004}}$	**n** $\mathbf{3\bar{1}4\bar{1}}$	**o** $\mathbf{2\bar{2}5\bar{1}}$ **or** $\mathbf{2\overline{151}}$	**p** $\mathbf{1\bar{3}}$

"And, in some very important and striking cases, sums requiring 30, 50, 100 or even more numerous and cumbrous "steps" of working (according to the current Western methods) can be answered in a single and simple step of work by the Vedic method! And little children (of only 10 or 12 years of age) merely look at the sums written on the blackboard (on the platform) and immediately shout out and dictate the answers from the body of the convocation hall (or other venue of demonstration). And this is because, as a matter of fact, each digit automatically yields its predecessor and its successor! and the children have merely to go on tossing off (or reeling off) the digits one after another (forwards or backwards) by mere mental arithmetic (without needing pen or pencil, paper or slate etc)!"

From "Vedic Mathematics", Page xvii.

9.4 USING BAR NUMBERS

Finally here are a few examples showing where bar numbers might be used.

13 **29 + 48 = 77**.

Writing 29 as $3\bar{1}$, or 48 as $5\bar{2}$:

$$\begin{array}{r} 29 \\ \underline{5\bar{2}} \; + \\ \underline{77} \end{array} \qquad \begin{array}{r} 3\bar{1} \\ \underline{48} \; + \\ \underline{77} \end{array}$$

14 **623 – 188 = 435**.

$$\begin{array}{r} 623 \\ \underline{2\bar{1}\bar{2}} \; - \\ \underline{435} \end{array}$$

15 **5032 + 7489 – 2883** = $10\bar{4}38$ = **9638**. We just add up the first digits of the first and second numbers and subtract the first digit of the third number. Similarly with the second, third and fourth digits.

16 **29 × 3 =** $3\bar{1} \times 3 = 9\bar{3}$ = **87**.

17 **87 ÷ 3 =** $9\bar{3} \div 3 = 3\bar{1}$ = **29**.

18 **41 ÷ 7 = 6 remainder** $\bar{1}$.

These bar numbers can be very useful in more advanced work (see Manuals 2 and 3).